

NHS Guidance Assistant — Design & Governance Note

A retrieval-augmented assistant over public NHS guidance — prototype built by Sobin Sunny

What it does

A working prototype that answers plain-English questions about NHS guidance documents. It retrieves the most relevant passages from a trusted document set and has a local language model answer **using only those passages**, showing the source of every answer. It runs entirely on-device, so no data leaves the machine — a deliberate privacy-first choice for public, potentially sensitive data.

Why this approach (and its limits)

Retrieval-Augmented Generation (RAG) is the right pattern here because the value is in **trusted, current, citable** answers, not in a model's general knowledge. Grounding every answer in retrieved source text is the single biggest safeguard against a model inventing information people might act on. The prototype is small by design; the note below sets out how I would take it to production responsibly.

How I would productionise it at NHSBSA

- **Cloud & MLOps:** containerise with Docker; deploy on the approved cloud (e.g. Azure) with infrastructure defined as code (Terraform); track experiments and model versions in MLflow so every change is reproducible and auditable.
- **Pipelines & CI/CD:** automated ingestion when guidance is updated, with tests and a human review gate before anything reaches users; version both the documents and the index.
- **Retrieval quality:** evaluate answers against a curated question set; monitor for retrieval drift and unanswered questions; tune chunking and top-k with evidence, not guesswork.
- **Scale:** swap the local FAISS index for a managed vector store (e.g. Pinecone/Milvus) and the local model for an approved, governed model endpoint as volumes grow.

Responsible AI, safety and the law

- **Grounding & honesty:** the assistant answers only from source guidance and explicitly says when an answer is not present — it does not guess.
- **Transparency & explainability:** every answer displays its source document and page, so any output can be checked and challenged.
- **Fairness & bias:** test across question phrasings and user groups; monitor for gaps that could disadvantage particular people; keep a human in the loop for anything consequential.
- **Data protection, FOI & equalities:** privacy-by-design (local processing, no data egress), clear records of sources and decisions, and alignment with ISO/IEC 42001 AI management practice for governance and accountability.

The stack, in one line

Python · LangChain · sentence-transformer embeddings · FAISS vector store · local open model via Ollama · Streamlit and CLI interfaces — with a clean, testable separation between configuration, ingestion, the RAG engine, and the UI.

Prototype built to demonstrate hands-on capability against the Senior AI/ML Engineer person specification. Happy to walk through the code and design decisions in person.